

# CS765 Spring 2023 Semester, Project Part-2

## Simulating a double selfish mining attack using P2P Cryptocurrency Network

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### **Question: Find the effect of the different adversary mining powers on these below ratios.**

To analyze the effect of different adversary mining powers on various ratios in a network, we need to consider parameters such as the number of peers, percentage of slow peers, percentage of lowCPU peers, mean transaction time, mean block generation time, and propagation delay. These parameters influence the effectiveness of selfish mining strategies and network stability.

Let's denote the parameters as follows:

- Number of Peers:  $N$
- Percentage of Slow Peers:  $\text{Slow}\%$
- Percentage of LowCPU Peers:  $\text{LowCPU}\%$
- Mean Transaction Time:  $T_{tx}$
- Mean Block Generation Time:  $T_{block}$
- Propagation Delay:  $\text{Delay}$

### **We are interested in the following ratios:**

1.  $MP_{U_{advnode}}$ : The ratio of the number of blocks mined by the adversary in the final public main chain to the total number of blocks mined by this adversary overall.
2.  $MP_{U_{overall}}$ : The ratio of the number of blocks in the final public main chain to the total number of blocks generated across all the nodes.

3.  $R_{pooladv}$ : The ratio of the number of blocks mined by the adversary in the final public main chain to the total number of blocks generated across all the nodes.
4.  $Fractionadv$ : The ratio of the number of blocks mined by the adversary in the final public main chain to the total number of blocks in the final public main chain.

## Effect of Parameters:

- **Number of Peers:** Increasing the number of peers generally reduces propagation delay, thus decreasing the likelihood of forks due to delayed block propagation.
- **Mean Transaction Time ( $T_{tx}$ ):** Shorter transaction times reduce the window of opportunity for selfish miners to execute attacks.
- **Mean Block Generation Time ( $T_{block}$ ):** Shorter block generation times make it harder for selfish miners to maintain hidden forks, as blocks are produced more frequently.
- **Propagation Delay:** Longer propagation times increase the probability of temporary forks in the blockchain, which can be exploited by selfish miners.

## Effect of Number of Peers in Network:

The number of peers in a network influences the speed at which new blocks and transactions propagate. Generally, increasing the number of peers reduces propagation delay, which decreases the likelihood of forks due to delayed block propagation. However, network latency can still play a significant role. If peers are located far from each other or if there are network congestion issues, propagation delay can increase regardless of the number of peers.

## Effect of Mean Transaction Time ( $T_{tx}$ ) in Network:

- **Short  $T_{tx}$ :** Short transaction times allow for faster confirmation and block addition. Because they have less time to hold blocks or launch double-spending operations, self-centered miners find it more difficult to carry out attacks. A shorter  $T_{tx}$  minimizes the window of opportunity when self-centered miners might outbid honest miners.
- **Long  $T_{tx}$ :** Transactions take longer to confirm and get added to blocks if the mean transaction time is high. In certain situations, selfish miners

may have a longer window of opportunity to carry out double-spending attacks or withhold blocks, giving them a larger chance to complete their attack. Selfish miners are more likely to successfully execute their attacks and make money from them if  $T_{tx}$  is longer.

## Effect of Mean Block Generation Time ( $T_{block}$ ) in Network:

- **Shorter  $T_{block}$ :** Blocks are being generated more often if the mean block time for generation is lower. This raises the probability that the blocks of selfish miners will become orphaned when they try to postpone the split and makes it more difficult for them to maintain a concealed fork of the network.
- **Longer  $T_{block}$ :** Blocks are created more rarely if the average block time for generation is longer. Because they have more time, selfish miners can create a longer private fork in a blockchain without the network as a whole understanding. This makes it more likely that self-centered miners will succeed in their attack by delaying the release of blocks to the server until their branch is longer than the main chain.
- **Impact on Selfish Mining Strategy:** Selfish miners' technique is directly affected by the average block creation time. Selfish miners may find it less beneficial to withhold blocks in networks with smaller block generation durations since the faster block creation rate increases the probability that honest miners will detect them. Since they have more time to attempt a private fork extension without being noticed, self-centered miners may find it more profitable to do so in networks with longer creation of blocks durations.

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## Selfish Mining Attack:

The selfish mining attack is a tactic used in blockchain networks, mainly seen in proof-of-work (PoW) based systems like Bitcoin. It involves a miner or a group of miners collaborating to gain an unfair advantage over honest miners by manipulating the consensus protocol.

1. **Private Block Mining:** To start, attackers mine blocks in private without disclosing them to the network. By doing this, they are able to generate a secret chain that is hidden from view by other miners.
2. **Monitoring the Public Chain:** Attackers monitor the public blockchain while mining in secret. Attackers compare a block obtained and broadcast by other miners with their own hidden chain.

3. **Strategic Release of Blocks:** If attackers discover that their hidden chain is longer than the public one, they strategically release their private blocks to the network. This aims to replace the public chain with their longer, hidden one, causing a reorganization of the blockchain.
4. **Orphaning Honest Blocks:** When attackers successfully replace the public chain with their hidden one, blocks mined by honest miners on the public chain become invalid. These discarded blocks waste the efforts of honest miners.

## Double Selfish Mining Attack:

The double selfish mining attack involves two distinct groups of miners independently executing selfish mining strategies without awareness of each other's actions. These groups work in parallel to exploit delays in block propagation across the network, aiming to replace the public blockchain with their longer private chains. This coordinated effort enhances their chances of successfully executing the attack, potentially allowing them to gain control over the blockchain and manipulate its transactions for their benefit.

## RESULTS OF SIMULATION

Here, in the Simulation 1 : Hashing Power of Adversary 2 is 0

| Hashing Power of Adversary 1 | Hashing Power of Adversary 2 | MPU of Adversary 1 | MPU of Adversary 2 | MPU Overall | Fraction of Adversary 1 | Fraction of Adversary 2 |
|------------------------------|------------------------------|--------------------|--------------------|-------------|-------------------------|-------------------------|
| 0                            | 0                            | 0                  | 0                  | 1           | 0                       | 0                       |
| 0.1                          | 0                            | 0.778              | 0                  | 0.846       | 0.2121                  | 0                       |
| 0.2                          | 0                            | 0.75               | 0                  | 0.8         | 0.25                    | 0                       |
| 0.3                          | 0                            | 0.667              | 0                  | 0.8048      | 0.2424                  | 0                       |
| 0.4                          | 0                            | 0.875              | 0                  | 0.74358     | 0.48275                 | 0                       |
| 0.5                          | 0                            | 0.722              | 0                  | 0.76        | 0.3421                  | 0                       |

Table 1: Hashing Power and related metrics for Adversaries

## EXPLANATION OF GRAPH

| Hashing Power of Adversary 1 | Hashing Power of Adversary 2 | MPU of Adversary 1 | MPU of Adversary 2 | MPU Overall | Fraction of Adversary 1 | Fraction of Adversary 2 |
|------------------------------|------------------------------|--------------------|--------------------|-------------|-------------------------|-------------------------|
| 0                            | 0                            | 0                  | 0                  | 1           | 0                       | 0                       |
| 0.1                          | 0                            | 0.778              | 0                  | 0.846       | 0.2121                  | 0                       |
| 0.2                          | 0                            | 0.75               | 0                  | 0.8         | 0.25                    | 0                       |
| 0.3                          | 0                            | 0.667              | 0                  | 0.8048      | 0.2424                  | 0                       |
| 0.4                          | 0                            | 0.875              | 0                  | 0.74358     | 0.48275                 | 0                       |
| 0.5                          | 0                            | 0.722              | 0                  | 0.76        | 0.3421                  | 0                       |

Table 2: Hashing Power and related metrics for Adversaries

## EXPLANATION OF GRAPH

| Hashing Power of Adversary 1 | Hashing Power of Adversary 2 | MPU of Adversary 1 | MPU of Adversary 2 | MPU Overall | Fraction of Adversary 1 | Fraction of Adversary 2 |
|------------------------------|------------------------------|--------------------|--------------------|-------------|-------------------------|-------------------------|
| 0                            | 0                            | 0                  | 0                  | 1           | 0                       | 0                       |
| 0.1                          | 0                            | 0.778              | 0                  | 0.846       | 0.2121                  | 0                       |
| 0.2                          | 0                            | 0.75               | 0                  | 0.8         | 0.25                    | 0                       |
| 0.3                          | 0                            | 0.667              | 0                  | 0.8048      | 0.2424                  | 0                       |
| 0.4                          | 0                            | 0.875              | 0                  | 0.74358     | 0.48275                 | 0                       |
| 0.5                          | 0                            | 0.722              | 0                  | 0.76        | 0.3421                  | 0                       |

Table 3: Hashing Power and related metrics for Adversaries